Non-parametric Tests

1. Mann-Whitney U Test

For the first test, I used a Mann-Whitney U test because the data is ratings from 1 to 10, which are ordinal. The two groups are independent because some people were assigned to the hairy spider and others to the non-hairy spider groups. The null hypothesis is that people do not rate hairy spiders as scarier than non-hairy spiders. The alternative hypothesis is that hairy spiders are rated as scarier than non-hairy spiders.

The Mann-Whitney U test gave a test statistic of U = 91.5 and a p-value of p = 0.0009. Since this p-value is less than 0.05, I reject the null hypothesis. This suggests that people rated hairy spiders as significantly scarier than non-hairy spiders.

```
1. Mann–Whitney U Test
U statistic: 91.5
p-value: 0.0008810296003726713
```

2. Fisher Exact Probability Test

For the second test I used Fisher's Exact Test because the data are categorical and arranged in a 2 by 2 table comparing ranking group and game outcome. Fisher's Exact Test is appropriate here because the sample size is small, so a chi-square test may not be as reliable. The null hypothesis is that winning is independent of League of Legends ranking. The alternative hypothesis is that winning is associated with ranking.

The Fisher Exact Test gave an odds ratio of 30.0 and a p-value of p = 0.0291. Since this p-value is less than 0.05, I reject the null hypothesis. This suggests that winning is not independent of ranking. In this data set, elite players were more likely to win than skilled players.

```
2. Fisher Exact Probability Test
Odds ratio: 30.0
p-value: 0.02913752913752914
```

3. Chi-square Test for Independence

For the third test I used a chi-square test for independence because the data are categorical and organized by two variables type of music and whether the player won or lost the game. The null hypothesis is that music type and game outcome are independent. The alternative hypothesis is that music type and game outcome are associated.

The chi-square test gave a chi-square statistic of $\chi^2 = 11.2540$, with 2 degrees of freedom, and a p-value of p = 0.0036. Since this p-value is less than 0.05 I reject the null hypothesis. This suggests that the type of music a person listened to was associated with whether they won or

lost the League of Legends match. Based on the observed data, players who listened to classical music had more wins while players who listened to metal had more losses.

```
3. Chi-square Test for Independence
Chi-square statistic: 11.254010695187166
p-value: 0.0035993379702155266
Degrees of freedom: 2
Expected frequencies:
[[ 9.73770492  6.19672131 11.06557377]
 [12.26229508  7.80327869 13.93442623]]
```

4. Wilcoxon Signed-Rank Test

For the fourth test I used a Wilcoxon signed-rank test because the data compare the same individuals before and after receiving the Covid vaccine. This makes the data paired, and the Wilcoxon test is appropriate when comparing paired data without assuming the differences are normally distributed. The null hypothesis is that the number of social interactions did not increase after vaccination. The alternative hypothesis is that the number of social interactions increased after vaccination.

The Wilcoxon signed-rank test gave a test statistic of W = 53.5 and a p-value of p = 0.0029. Since this p-value is less than 0.05, I reject the null hypothesis. This suggests that individuals had significantly more social interactions after receiving the Covid vaccine.

```
4. Wilcoxon Signed-Rank Test
W statistic: 53.5
p-value: 0.0029296875
```